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Lecture 08: Dynamic OT Part I

1 Motivation

- Static OT: works with couplings, $\pi(x, y)$
- Dynamic OT: studies the **time evolution** of distributions

Dynamic OT has applications in Physics (mass flow, fluid mechanics), Probability (diffusion, stochastic control), and ML (trajectory alignment, generative models).

Static OT

- Considers couplings transporting $\alpha \to \beta$
- Transport is one-shot, static
- Find coupling minimizing distance
- **Key Idea:** OT as a min-cost (soft) matching between distributions

Dynamic OT

- Considers flows ρ_t, v_t transporting α, β
- Transport is time-dependent, dynamic
- Find flow minimizing energy
- **Key Idea:** OT as shortest path (geodesic) in probability space

2 Benamou-Brenier Formulation

Measures α, β, cost$=\|x-y\|_2^2$

Definition 1 (Benamou-Brenier Formulation)**.**

$$W_2^2(\alpha,\beta) = \inf_{\rho_t, v_t} \int_0^1 \int_{\mathcal{X}} \overbrace{\|v_t(x)\|^2}^{\text{kinetic energy (speed}^2\text{) of particle at location } x} \overbrace{\rho_t(x)}^{\text{kinetic energy per unit volume transported mass}} dx dt, \tag{1}$$

subject to boundary conditions

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \qquad \textit{continuity equation} \tag{2}$$

$$\rho_0 = \alpha, \qquad \textit{initial condition} \tag{3}$$

$$\rho_1 = \beta, \qquad \textit{ending condition} \tag{4}$$

where ρ_t is a measure and v_t is a vector field at time t. The integral over $\mathcal{X}$ reprsents the total ρ-weighted kinetic energy at time t and the integral over t represents the total energy needd over the entire transport process.

2.1 Continuity Equation

The continuity equation

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0, \tag{5}$$

ensures the system obeys physical constraints. $\partial_t \rho_t$ measures the change in density ρ at x, and $\nabla \cdot (\rho_t v_t)$ represents the divergence of flux $\rho_t v_t$, or the net flow of mass entering and leaving x. The continuity equation is a **mass conservation constraint** that ensures probability mass is neither created nor destroyed. The divergence of the flow into/out of point x should equal the change in density at that point x.

3 Geodesics in Wasserstein Space

B.B. shows W_2 is a geodesic in $\mathcal{P}(\mathcal{X})$. A geodesic is the shortest constant speed path contecting two points in space. A geodesic distance is one where distance = length of the shortest path. Thus, W_2 is a natural way to encode the geometry of the space of distributions! Furthermore, interpolation can be viewed as evolving densities along minimal-energy paths.

3.0.1 McCann's Interpolation

Definition 2 (Displacement Interpolation, or McCann's Interpolation)**.** *If an OT map T exists between α to β, the optimal path is given by*

$$\rho_t = [(1-t)\mathrm{Id} + tT]_{\#}\alpha. \tag{6}$$

The corresponding vector field is

$$v_t(x) = T(x) - x. \tag{7}$$

For a given point x, we can map its trajectory via the (naive) interpolation

$$x_t = (1-t)x + t(T(x)). \tag{8}$$

Remark: This looks like linear interpolation but it is interpolation in the probability space.

The curve $(\rho_t)_{t\in[0,1]}$ is a geodesic in Wasserstein space.
If no Monge map exists, we can still define displacement with the Kantorovich solution, or coupling π:

$$\rho_t = [(1-t)x + ty]_{\#}\pi \tag{9}$$

What about naively interpolating between measures?

Definition 3 (Mixture Interpolation)**.**

$$\rho_t^{\text{mix}} = (1-t)\alpha + t\beta \tag{10}$$

Displacement interpolation interpolates between **locations** of points whereas mixture interpolation interpolates **masses** of points. Mixture is easy to compute but is **not** a geodesic in $\mathbb{W}_p$.

4 Geometric Properties

$\mathbb{W}_2$ is a Riemannian manifold, *nonnegative curvature*.
Convexity of functionals along Wasserstein geodesics provides a notion of curvature.

Definition 4 (Euclidean Convexity)**.** *A functional $F : \mathbb{R}^d \to \mathbb{R}$ is convex if:*

$$F((1-t)x_0 + tx_1) \le (1-t)F(x_0) + tF(x_1), t \in [0,1]. \tag{11}$$

Convexity along straight lines encodes flatness.

Definition 5 (Wasserstein Convexity)**.** *A functional $F : \mathbb{W}_2(\mathbb{R}^d) \to \mathbb{R}$ is **displacement convex** if:*

$$F(\rho_t) \le (1-t)F(\rho_0) + tF(\rho_1), t \in [0,1]. \tag{12}$$

References